

Roll. No.

O.M.R. Serial No.

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Question Booklet Number

200002

B.C.A. (SEM.-II) (NON-NEP) EXAMINATION, 2024

COMPUTER APPLICATION

(C Programming)

[BCA-201(N)]

Paper Code

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Question Booklet
Series

B

Time : 1 : 30 Hours

Max. Marks : 75

Instructions to the Examinee :

1. Do not open the booklet unless you are asked to do so.
2. The booklet contains 100 questions. Examinee is required to answer 75 questions in the OMR Answer-Sheet provided and not in the question booklet. All questions carry equal marks.
3. Examine the Booklet and the OMR Answer-Sheet very carefully before you proceed. Faulty question booklet due to missing or duplicate pages/questions or having any other discrepancy should be got immediately replaced.
4. Four alternative answers are mentioned for each question as - A, B, C & D in the booklet. The candidate has to choose the correct / answer and mark the same in the OMR Answer-Sheet as per the direction :

(Remaining instructions on last page)

परीक्षार्थियों के लिए निर्देश :

1. प्रश्न-पुस्तिका को तब तक न खोलें जब तक आपसे कहा न जाए।
2. प्रश्न-पुस्तिका में 100 प्रश्न हैं। परीक्षार्थी को 75 प्रश्नों को केवल दी गई OMR आन्सर-शीट पर ही हल करना है, प्रश्न-पुस्तिका पर नहीं। सभी प्रश्नों के अंक समान हैं।
3. प्रश्नों के उत्तर अंकित करने से पूर्व प्रश्न-पुस्तिका तथा OMR आन्सर-शीट को सावधानीपूर्वक देख लें। दोषपूर्ण प्रश्न-पुस्तिका जिसमें कुछ भाग छपने से छूट गए हों या प्रश्न एक से अधिक बार छप गए हों या उसमें किसी अन्य प्रकार की कमी हो, उसे तुरन्त बदल लें।
4. प्रश्न-पुस्तिका में प्रत्येक प्रश्न के चार सम्भावित उत्तर- A, B, C एवं D हैं। परीक्षार्थी को उन चारों विकल्पों में से सही उत्तर छँटना है। उत्तर को OMR उत्तर-पत्रक में सम्बन्धित प्रश्न संख्या में निम्न प्रकार भरना है :

(शेष निर्देश अन्तिम पृष्ठ पर)

S

1. Which of the following statements is true about `realloc()`?
 - (A) It can only increase the size of memory
 - (B) It can only decrease the size of memory
 - (C) It can both increase and decrease the size of memory
 - (D) It cannot resize memory
2. What is the return type of `malloc()`, `calloc()`, and `realloc()` functions?
 - (A) void
 - (B) int
 - (C) char
 - (D) Pointer
3. Which of the following statements about dynamic memory allocation is false?
 - (A) Dynamic memory allocation allows for flexibility in memory usage.
 - (B) Memory allocated dynamically persists until the end of the program.
 - (C) Dynamic memory allocation requires manual deallocation using `free()`.
 - (D) Dynamic memory allocation can lead to memory leaks if not handled properly.
4. Which of the following functions can be used to allocate memory for an array of structures?
 - (A) `malloc()`
 - (B) `calloc()`
 - (C) `realloc()`
 - (D) All of the above
5. In dynamic memory allocation, which function is used to allocate memory for an array of elements with initialization?
 - (A) `malloc()`
 - (B) `calloc()`
 - (C) `realloc()`
 - (D) `initmem()`

6. What is the purpose of dynamic memory allocation in C?
- (A) To allocate memory at compile time
 - (B) To allocate memory at runtime
 - (C) To deallocate memory at compile time
 - (D) To deallocate memory at runtime
7. Which library function is used to allocate memory dynamically in C?
- (A) malloc()
 - (B) calloc()
 - (C) realloc()
 - (D) All of the above
8. What does the malloc() function return if it fails to allocate memory?
- (A) NULL
 - (B) 0
 - (C) -1
 - (D) Undefined behavior
9. How do you deallocate memory allocated by malloc() in C?
- (A) Using free()
 - (B) Using delete
 - (C) Using release()
 - (D) Using dealloc()
10. What is the difference between malloc() and calloc() functions?
- (A) malloc() initializes memory to 0, calloc() does not
 - (B) malloc() initializes memory to 1, calloc() initializes memory to 0
 - (C) malloc() allocates memory for a single element, calloc() allocates memory for multiple elements
 - (D) There is no difference
11. Which function is used to resize dynamically allocated memory in C?
- (A) realloc()
 - (B) resize()
 - (C) expand()
 - (D) modify()

12. What is the purpose of the `strstr()` function in C?
- (A) To search for a substring within a string
 - (B) To compare two strings
 - (C) To concatenate two strings
 - (D) To find the length of a string
13. What is the output of the following code snippet?
- ```
char str1[] = "Hello"; char str2[] = "World"; printf("%s", strcat(str1, str2));
```
- (A) "Hello"
  - (B) "World"
  - (C) "HelloWorld"
  - (D) Compiler error
14. Which function is used to convert a string to an integer in C?
- (A) `atoi()`
  - (B) `tostring()`
  - (C) `itoa()`
  - (D) `strint()`
15. Which header file is used for file handling in C?
- (A) `<stdio.h>`
  - (B) `<stdlib.h>`
  - (C) `<file.h>`
  - (D) `<io.h>`
16. What is the purpose of `fopen()` function in C?
- (A) To close a file
  - (B) To open a file
  - (C) To read from a file
  - (D) To write to a file
17. Which mode in `fopen()` is used to open a file for writing, starting from the beginning of the file, and truncating its contents?
- (A) "r"
  - (B) "w"
  - (C) "a"
  - (D) "r+"
18. Which function is used to close a file in C?
- (A) `fclose()`
  - (B) `close()`
  - (C) `fcloseall()`
  - (D) `closeall()`

19. What is the purpose of the void pointer in C?
- (A) To declare a pointer variable without specifying its data type
  - (B) To access memory addresses
  - (C) To perform arithmetic operations on pointers
  - (D) To perform bitwise operations
20. Which of the following statements about pointers in C is false?
- (A) Pointers can be used to implement dynamic memory allocation.
  - (B) Pointers can be used to access elements of an array.
  - (C) Pointers can only store addresses of variables allocated on the heap.
  - (D) Pointers can be used to implement data structures like linked lists.
21. What is a string in C?
- (A) A data type to store a single character
  - (B) A data type to store integers
  - (C) A sequence of characters terminated by a null character
  - (D) A reserved keyword
22. Which header file must be included to work with strings in C?
- (A) <stdio.h>
  - (B) <stdlib.h>
  - (C) <string.h>
  - (D) <math.h>
23. What is the null character in C?
- (A) '\0'
  - (B) '\n'
  - (C) '\t'
  - (D) '\a'
24. What is the maximum length of a string in C?
- (A) 255 characters
  - (B) 1024 characters
  - (C) No fixed limit
  - (D) 100 characters

25. What is the correct way to access a member "b" of a structure "B" nested within another structure "A"?
- (A) A.b  
(B) A.B.b  
(C) A->B->b  
(D) A.B->b
26. What is the correct syntax to define a structure pointer variable named "ptr" that points to a structure of type "Book"?
- (A) struct Book ptr;  
(B) Book ptr;  
(C) struct \*ptr;  
(D) Book \*ptr;
27. What is a pointer in C?
- (A) A data type used to store integers  
(B) A variable that holds the address of another variable  
(C) A reserved keyword  
(D) A loop construct
28. What symbol is used to declare a pointer variable in C?
- (A) &  
(B) \*
- (C) \$  
(D) # \*\*
29. What is the purpose of the ampersand (&) operator in C?
- (A) To declare a pointer variable  
(B) To access the value stored at a memory address  
(C) To assign a value to a pointer variable  
(D) To obtain the address of a variable
30. Which of the following statements is true about pointers in C?
- (A) Pointers cannot be used with arrays.  
(B) Pointers can only store addresses of variables of the same data type.  
(C) Pointers cannot be dereferenced.  
(D) Pointers can be assigned NULL to indicate that they are not pointing to any valid address.

Answer:

31. Which of the following statements is true regarding arrays in C?
- (A) Arrays can only store primitive data types
  - (B) Arrays can store both primitive and user-defined data types
  - (C) Arrays cannot store strings
  - (D) Arrays can only store integers
32. How do you pass an array to a function in C?
- (A) By passing each element separately
  - (B) By passing the array's address
  - (C) By passing the array's size
  - (D) By passing a pointer to the array
33. What is the value of `arr[3]` if `int arr[5] = {10, 20, 30, 40, 50};`?
- (A) 10
  - (B) 20
  - (C) 30
  - (D) 40
34. What is the name of the memory location of the first element of an array?
- (A) Array
  - (B) Index
  - (C) Base address
  - (D) Pointer
35. Which of the following is not a valid way to initialize an array in C?
- (A) `int arr[] = {1, 2, 3};`
  - (B) `int arr[3] = {1, 2};`
  - (C) `int arr[3] = {1};`
  - (D) `int arr[3] = {1, 2, 3, 4};`
36. What happens if you try to access an element of an array beyond its bounds in C?
- (A) Compile-time error
  - (B) Runtime error
  - (C) No error, the program continues execution
  - (D) Warning message

37. Which of the following statements about structures in C is true?
- (A) Structures cannot contain arrays as members.
  - (B) Structures cannot be passed to functions.
  - (C) Structures cannot be nested.
  - (D) Structures can contain other structures as members.
38. How do you access members of a structure in C?
- (A) By using dot notation
  - (B) By using parentheses
  - (C) By using square brackets
  - (D) By using curly braces
39. What is the purpose of typedef in C when used with structures?
- (A) To define a new structure type
  - (B) To declare structure members
  - (C) To initialize structures
  - (D) To define structure variables
40. How do you initialize a structure variable in C?
- (A) By using the init keyword
  - (B) By using square brackets
  - (C) By using curly braces
  - (D) By using parentheses
41. Which of the following statements is true regarding structures in C?
- (A) Structures cannot contain functions as members.
  - (B) Structures can only contain variables of the same data type.
  - (C) Structures cannot be returned from functions.
  - (D) Structures cannot be assigned to other structures.
42. What is the size of a structure in C?
- (A) The sum of sizes of its members
  - (B) The average size of its members
  - (C) The size of its largest member
  - (D) The size of its smallest member

43. How is a function called in C?
- (A) By using its return type
  - (B) By using its name followed by parentheses containing arguments (if any)
  - (C) By using the keyword "call"
  - (D) None of the above
44. What is a recursive function?
- (A) A function that calls itself
  - (B) A function that is called by another function
  - (C) A function that returns void
  - (D) A function that returns a pointer
45. What is the maximum number of functions that can be called from main() in C?
- (A) 256
  - (B) 127
  - (C) No limit
  - (D) 32
46. What is the purpose of the return statement in a function?
- (A) To terminate the function
  - (B) To return a value to the calling function
  - (C) To declare the return type of the function
  - (D) None of the above
47. What happens if a function is called before it is declared or defined in C?
- (A) Compilation error
  - (B) Runtime error
  - (C) Warning message
  - (D) No effect
48. What is the significance of the static keyword in a function declaration in C?
- (A) It makes the function private
  - (B) It makes the function public
  - (C) It makes the function static
  - (D) It makes the function external
49. How are arguments passed to a function in C?
- (A) By value
  - (B) By reference
  - (C) By pointer
  - (D) All of the above

50. What is the syntax of `realloc()` function in C?
- (A) `ptr = realloc(ptr, new_size);`
  - (B) `ptr = realloc(new_size, ptr);`
  - (C) `realloc(ptr, new_size);`
  - (D) `realloc(new_size, ptr);`
51. What happens if `realloc()` fails to resize memory?
- (A) Returns NULL
  - (B) Returns the original pointer
  - (C) Returns 0
  - (D) Exits the program
52. What does the `calloc()` function do before returning a pointer to allocated memory?
- (A) Initializes memory to 0
  - (B) Initializes memory to random values
  - (C) Does not initialize memory
  - (D) Frees memory
53. How do you allocate an array dynamically using `malloc()`?
- (A) `int *arr = malloc(sizeof(int) * n);`
  - (B) `int arr[] = malloc(n * sizeof(int));`
  - (C) `int arr[n];`
  - (D) `int arr = malloc(n * sizeof(int));`
54. What is a memory leak in C?
- (A) When memory is deallocated correctly
  - (B) When memory is allocated but never deallocated
  - (C) When memory is allocated and deallocated excessively
  - (D) When memory is allocated statically

55. What is the purpose of `fseek()` function in C?
- (A) To close a file
  - (B) To move the file pointer to a specified position
  - (C) To read from a file
  - (D) To write to a file
56. Which function is used to read data from a file in C?
- (A) `fread()`
  - (B) `read()`
  - (C) `fgets()`
  - (D) `scanf()`
57. Which of the following modes in `fopen()` is used to open a file for appending data?
- (A) "r"
  - (B) "w"
  - (C) "a"
  - (D) "r+"
58. What is the purpose of `fprintf()` function in C?
- (A) To close a file
  - (B) To read data from a file
  - (C) To write formatted data to a file
  - (D) To move the file pointer to a specified position
59. Which function is used to delete a file in C?
- (A) `delete()`
  - (B) `unlink()`
  - (C) `remove()`
  - (D) `erase()`
60. Which of the following is true about binary files in C?
- (A) Binary files are human-readable.
  - (B) Binary files can only be opened in "r" mode.
  - (C) Binary files store data in a format that is not human-readable.
  - (D) Binary files cannot be read or written using standard library.

61. What is the correct syntax to declare a pointer to an integer variable in C?
- (A) `int ptr;`
  - (B) `ptr int;`
  - (C) `int *ptr;`
  - (D) `*int ptr;`
62. What is a NULL pointer in C?
- (A) A pointer that stores the value 0
  - (B) A pointer that points to a non-existent memory location
  - (C) A pointer that is uninitialized
  - (D) A reserved keyword
63. What is the purpose of the arrow (`->`) operator in C?
- (A) To declare a pointer variable
  - (B) To access the value stored at a memory address
  - (C) To access members of a structure through a pointer
  - (D) To perform bitwise operations
64. What is pointer arithmetic in C?
- (A) Arithmetic operations performed on pointer variables
  - (B) Performing arithmetic operations using the address of a variable
  - (C) A special type of loop construct
  - (D) A reserved keyword
65. Which of the following is the correct syntax to increment a pointer in C?
- (A) `ptr++`
  - (B) `++ptr`
  - (C) `ptr += 1`
  - (D) All of the above
66. What is the size of a pointer variable in C?
- (A) 1 byte
  - (B) 2 bytes
  - (C) 4 bytes
  - (D) Depends on the system architecture
67. Which of the following statements about pointers in C is true?
- (A) Pointers can only store addresses of variables allocated on the stack.
  - (B) Pointers can be compared using relational operators.
  - (C) Pointers cannot be assigned to another pointer.
  - (D) Pointers are automatically initialized to NULL.

68. What is the correct way to access the “age” member of a structure variable named “person”?
- (A) person.age
  - (B) person->age
  - (C) person[age]
  - (D) person::age
69. What is the purpose of the arrow operator (->) in C when used with structures?
- (A) To access structure members using pointer variables
  - (B) To define a structure
  - (C) To declare a structure variable
  - (D) To initialize a structure
70. Which of the following statements is true regarding the initialization of structure members in C?
- (A) Only one member can be initialized at a time.
  - (B) All members must be initialized in the same statement.
  - (C) Members can be initialized in any order.
  - (D) Members cannot be initialized.
71. What is the purpose of the dot (.) operator in C when used with structures?
- (A) To define a structure
  - (B) To access structure members
  - (C) To declare a structure variable
  - (D) To initialize a structure
72. What is the correct way to declare a pointer to a structure in C?
- (A) ptr \*Person;
  - (B) ptr Person;
  - (C) struct \*ptr;
  - (D) struct ptr;
73. Which of the following statements is true regarding nested structures in C?
- (A) Nested structures are not allowed in C.
  - (B) Nested structures can only contain primitive data types.
  - (C) Nested structures cannot be accessed directly.
  - (D) Nested structures can have members of any data type.

74. Which function is used to calculate the length of a string in C?
- (A) strlen()
  - (B) strlength()
  - (C) length()
  - (D) size()
75. What is the purpose of the strcpy() function in C?
- (A) To concatenate two strings
  - (B) To compare two strings
  - (C) To copy one string to another
  - (D) To find the length of a string
76. Which of the following functions is used to concatenate two strings in C?
- (A) strcat()
  - (B) strconcat()
  - (C) concat()
  - (D) join()
77. What is the output of the following code snippet?
- ```
char str[] = "Hello"; printf("%c", str[4]);
```
- (A) 'H'
 - (B) 'e'
 - (C) 'o'
 - (D) Null character
78. Which of the following statements is true about strings in C?
- (A) Strings are mutable.
 - (B) Strings cannot contain spaces.
 - (C) Strings are terminated by the newline character.
 - (D) Strings can be manipulated using standard library functions.
79. What is the correct way to declare a string in C?
- (A) string str;
 - (B) char str[];
 - (C) str[] = "Hello";
 - (D) char str[10];
80. What function is used to find the first occurrence of a character in a string in C?
- (A) strchr()
 - (B) strfind()
 - (C) find()
 - (D) search()

81. What is the syntax for accessing the last element of an array `arr` of size `n`?
- (A) `arr[n-1]`
 - (B) `arr[n]`
 - (C) `arr[n+1]`
 - (D) `arr[n-2]`
82. Which of the following is true about multi-dimensional arrays in C?
- (A) They are limited to two dimensions
 - (B) They can have more than two dimensions
 - (C) They can only store integers
 - (D) They are stored in contiguous memory locations
83. How do you find the length of an array in C?
- (A) By using the `length` property
 - (B) By subtracting the starting address from the ending address
 - (C) By using the `sizeof` operator
 - (D) By using the `lengthof` function
84. What is a structure in C?
- (A) A function that performs a specific task
 - (B) A collection of variables of different data types under one name
 - (C) A reserved keyword
 - (D) A loop construct
85. What keyword is used to define a structure in C?
- (A) `struct`
 - (B) `structure`
 - (C) `def`
 - (D) None, structures are predefined in C
86. What is the size of an empty structure in C?
- (A) 0 bytes
 - (B) 1 byte
 - (C) 4 bytes
 - (D) Depends on the compiler

87. What is the default return type of a function in C if no return type is specified?
- (A) void
(B) int
(C) char
(D) float
88. What is an array in C?
- (A) A single variable that holds multiple values of different data types
(B) A collection of variables of the same data type that share the same name
(C) A built-in data type in C
(D) A reserved keyword
89. What is the index of the first element in an array in C?
- (A) 0
(B) 1
(C) -1
(D) Depends on the size of the array
90. What is the maximum number of elements that an array can hold in C?
- (A) 255
(B) 1024
(C) No fixed limit
(D) 100 t
91. How do you access elements of an array in C?
- (A) By using dot notation
(B) By using parentheses
(C) By using square brackets and index
(D) By using curly braces
92. What is the size of the following array declaration: `int arr[5];`?
- (A) 5
(B) 10
(C) 20
(D) Depends on the system architecture
93. Which of the following is the correct syntax to initialize an array in C?
- (A) `int arr[] = {1, 2, 3, 4, 5};`
(B) `int arr[5] = {1, 2, 3, 4, 5};`
(C) `int arr[5] = (1, 2, 3, 4, 5);`
(D) All of the above

94. What is a function in C?
- (A) A reserved word
 - (B) A set of statements grouped together to perform a specific task
 - (C) A data type
 - (D) A type of loop
95. What is the return type of a function in C?
- (A) void
 - (B) int
 - (C) char
 - (D) All of the above
96. What keyword is used to declare a function in C?
- (A) func
 - (B) function
 - (C) def
 - (D) None, functions are declared implicitly
97. Which of the following is not a valid function declaration in C?
- (A) `int add(int a, int b);`
 - (B) `void sum(int x, int y);`
 - (C) `float multiply(float p, float q)`
`{}`
 - (D) None of the above
98. What is the purpose of a function prototype in C?
- (A) To define the function body
 - (B) To declare the function's return type and parameters
 - (C) To include libraries
 - (D) None of the above
99. Which of the following statements about function parameters is true?
- (A) Parameters are optional in C functions
 - (B) Parameters cannot have default values
 - (C) Parameters are separated by commas in the function declaration
 - (D) All of the above
100. What is the scope of a variable declared inside a function in C?
- (A) Global
 - (B) Local to the function
 - (C) Local to the file
 - (D) Both ((A) and ((B)

ROUGH WORK

Example :

Question :

Q.1 (A) ● (C) (D)

Q.2 (A) (B) ● (D)

Q.3 (A) ● (C) (D)

5. Each question carries equal marks. Marks will be awarded according to the number of correct answers you have.
6. All answers are to be given on OMR Answer Sheet only. Answers given anywhere other than the place specified in the answer sheet will not be considered valid.
7. Before writing anything on the OMR Answer Sheet, all the instructions given in it should be read carefully.
3. After the completion of the examination, candidates should leave the examination hall only after providing their OMR Answer Sheet to the invigilator. Candidate can carry their Question Booklet.
9. There will be no negative marking.
10. Rough work, if any, should be done on the blank pages provided for the purpose in the booklet.
11. To bring and use of log-book, calculator, pager & cellular phone in examination hall is prohibited.
12. In case of any difference found in English and Hindi version of the question, the English version of the question will be held authentic.

Impt. On opening the question booklet, first check that all the pages of the question booklet are printed properly. If there is any discrepancy in the question Booklet, then after showing it to the invigilator, get another question Booklet of the same series.

उदाहरण :

प्रश्न :

प्रश्न 1 (A) ● (C) (D)

प्रश्न 2 (A) (B) ● (D)

प्रश्न 3 (A) ● (C) (D)

5. प्रत्येक प्रश्न के अंक समान हैं। आपके जितने उत्तर सही होंगे, उन्हीं के अनुसार अंक प्रदान किये जायेंगे।
6. सभी उत्तर केवल ओ०एम०आर० उत्तर-पत्रक (OMR Answer Sheet) पर ही दिये जाने हैं। उत्तर-पत्रक में निर्धारित स्थान के अलावा अन्यत्र कहीं पर दिया गया उत्तर मान्य नहीं होगा।
7. ओ०एम०आर० उत्तर-पत्रक (OMR Answer Sheet) पर कुछ भी लिखने से पूर्व उसमें दिये गये सभी अनुदेशों को सावधानीपूर्वक पढ़ लिया जाये।
8. परीक्षा समाप्ति के उपरान्त परीक्षार्थी कक्ष निरीक्षक को अपनी OMR Answer Sheet उपलब्ध कराने के बाद ही परीक्षा कक्ष से प्रस्थान करें। परीक्षार्थी अपने साथ प्रश्न-पुस्तिका ले जा सकते हैं।
9. निगेटिव मार्किंग नहीं है।
10. कोई भी रफ कार्य, प्रश्न-पुस्तिका में, रफ-कार्य के लिए दिए खाली पेज पर ही किया जाना चाहिए।
11. परीक्षा-कक्ष में लॉग-बुक, कैल्कुलेटर, पेजर तथा सेल्युलर फोन ले जाना तथा उसका उपयोग करना वर्जित है।
12. प्रश्न के हिन्दी एवं अंग्रेजी रूपान्तरण में भिन्नता होने की दशा में प्रश्न का अंग्रेजी रूपान्तरण ही मान्य होगा।

महत्वपूर्ण: प्रश्नपुस्तिका खोलने पर प्रथमतः जाँच कर देख लें कि प्रश्नपुस्तिका के सभी पृष्ठ भलीभाँति छपे हुए हैं। यदि प्रश्नपुस्तिका में कोई कमी हो, तो कक्षनिरीक्षक को दिखाकर उसी सिरीज की दूसरी प्रश्नपुस्तिका प्राप्त कर लें।

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